

Mellinger-Garrett_individual-assignment-05

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Set up

Packages

```
#|label: packages

# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
  "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

Cleaning sites object

Goal: create a new object from `sites` that only includes NCOS sites

Use: filter downstream data frames using new object

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

Cleaning taxon_list

Goal: convert character strings in existing `taxon_list` data frame into snake case

Use: join with aquatic inverts to attach taxonomic information to species IDs

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

Cleaning water_quality

Goal: create data frame with mean pH, mean DO (in mg/L), and mean salinity (in ppt) for each date and site

Use: join with aquatic inverts to attach water quality information to surveys

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning aq_ins

Goal: create long format data frame in which each row contains the counted individuals in each taxon during each survey on each date with all the taxonomic information for the taxon and the water quality information for the survey

Use: visualize relationships between water quality parameters and aquatic invertebrate abundances

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
```

```

# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter
# (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site

```

```
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median",
                              .na_rm = TRUE),
       site_full = fct_rev(site_full))
```

1. Visualizing differences across sites in major taxon abundance

a. Planning

- x-axis: site
- y-axis: log-transformed ostracod average abundance/L
- geometry to represent average abundance/L: points representing individual log-transformed average ostracod abundance/L values
- geometry to represent medians: larger points with a different color, shape, and size representing the median abundance/L at each site

b. Wrangling

```
#|label: wrangling ostracod data

# create new dataframe from aq_ins_clean
ostracod_df <- aq_ins_clean |>
  # keep only ostracod observations
  filter(taxon == "ostracod") |>
  # create column with log transformed abundance
  mutate(log_abundance = log(ave_lit + 1))

# display 10 random rows from data frame
ostracod_df |>
  slice_sample(n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-02-21_NEC	2023-02-21 00:00:00	NEC	ostr~	0	8.56	NA	8.8
2	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.2	NA	NA	NA
3	2021-08-25_NPB	2021-08-25 00:00:00	NPB	ostr~	581.	7.52	4.57	4.5
4	2023-11-11_NEC	2023-11-11 00:00:00	NEC	ostr~	0	NA	38.6	0.25

```

5 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR ostr~ 4.8 7.34 4.1 6.1
6 2022-11-19_NDC 2022-11-19 00:00:00 NDC ostr~ 6 NA NA NA
7 2023-11-12_NMC 2023-11-12 00:00:00 NMC ostr~ 0.533 NA 47 14.0
8 2023-09-15_NMC 2023-09-15 00:00:00 NMC ostr~ 0.4 NA NA 14.2
9 2022-05-11_NDC 2022-05-11 00:00:00 NDC ostr~ 0 9.11 0.398 9.4
10 2023-08-12_NEC 2023-08-12 00:00:00 NEC ostr~ 0 NA NA NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_abundance <dbl>

```

c. Coding

```

# base layer of ggplot as the ostracod dataframe we just created
ostracods <- ggplot(data = ostracod_df,
                    # x-axis: site
                    # y-axis: mean abundance/L
                    mapping = aes(x = site_full,
                                  y = log_abundance)) +

# points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +

# points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +

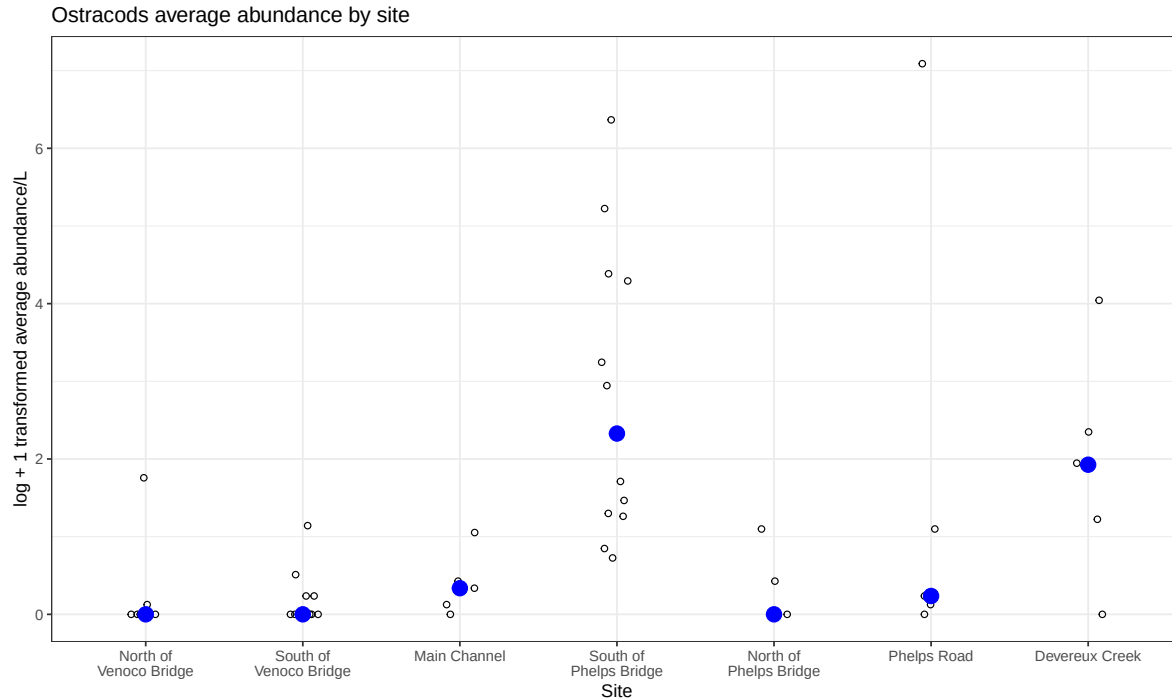
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +

# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods average abundance by site") +

# different theme
theme_bw()

# displaying object
ostracods

```



d. Creating multipanel figure

```
# base layer for salinity plot
salinity <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = site_full,
                                y = med_sal)) +

# points representing salinity observations
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

# points representing median salinity
stat_summary(fun = median,
            geom = "point",
            size = 4,
            color = "red") +

# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(15)) +

# relabelling axes and adding title
labs(x = "Site",
     y = "Median salinity (ppt)",
```

```

    title = "Site salinity") +
# changing theme
theme_bw()

# create copepod dataframe
copepod_df <- aq_ins_clean |>
  # keep only copepod observations
  filter(taxon == "copepod") |>
  # create log transformed abundance column
  mutate(log_abundance = log(ave_lit + 1))

# base layer of copepod plot
copepods <- ggplot(data = copepod_df,
                   mapping = aes(x = site_full,
                                y = log_abundance)) +
  # points representing observations
  geom_jitter(height = 0,
             shape = 21,
             width = 0.1) +
  # points representing medians
  stat_summary(geom = "point",
              fun = "median",
              size = 4,
              color = "green") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling axes and adding title
  labs(x = "Site",
       y = "log + 1 average abundance/L",
       title = "Copepods average abundance by site") +
  # changing theme
  theme_bw()

# combining plots into multipanel figure
salinity / (copepods | ostracods)

```